Response to reviewers

Anders Munch and Thomas A. Gerds

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We thank the reviewers for their useful comments. We have incorporated their suggestions into the manuscript, and we find it much improved. Please see our point-to-point responses below.

Quotes from the updated manuscript is inserted in gray boxes and when relevant newly added text is show in blue.

Reviewer 1

Comments to the Author

This paper proposes a super learner-style estimator of conditional cumulative hazard functions in the context of right-censored time-to-event data with competing risks. The user specifies libraries of candidate estimators of the conditional cumulative hazards of the event, censoring, and competing risk times. The proposed method then selects the element of the product of these libraries that has the smallest cross-validated empirical risk, where the risk is based on a carefully constructed loss function. A key benefit of the proposed methods over alternatives is that it jointly estimates the three cumulative hazards, rather than iterating between estimation of them or requiring a pre-specified censoring survival estimator. The authors provided an oracle inequality demonstrating that the true risk of the estimator is no worse than a constant multiple of the oracle risk up to a logarithmic penalty in the size of the library. The authors also provide numerical studies and a simulation illustrating the good performance of the proposed method.

Overall, this is a nice contribution to the literature, and I am grateful to the authors for the methods and accompanying theory. The paper is well written and careful, though I think more effort could be made in parts to make the notation somewhat more accessible. In particular, an outline of the proposed algorithm in its totality would be useful for readers.

Thank you for the encouraging words. We agree that it is a good idea with an outline of the proposed algorithm and we have added a pseudo-algorithm at the end of section 4.

1. I have two main comments. My first comment regards a sentence in the discussion: “A potential drawback of our approach is that we are evaluating the loss of the learners on the level of the observed data distribution, while the target of the analysis is either the event-time distribution, or the censoring distribution, or both.” I didn’t quite catch this on a first read-through of the paper, but I think it is an important point that deserves to be emphasized more throughout the paper. Specifically, this makes me wonder about the relevance of the oracle inequalities in Proposition 2 and Corollary 3: do we actually care about these risks? I usually care about inequalities and rates of convergence of the conditional survival (or cumulative hazard) function of the event time, censoring time, and competing risk. If I understand the above comment correctly, the risk considered in the paper doesn’t directly give such inequalities and rates. Is that correct? If so, it deserves to be mentioned and at least briefly discussed in the section on theoretical results. Similarly, this makes me question the relevance of the IPA metric considered in the numerical study. How do the different algorithms compare in terms of pointwise or integrated error for the conditional survival or cumulative hazard functions?

This is an important point, which we agree should be made clear earlier in the paper. We have now added a paragraph at the end of Section 5 (see below). We have also added a sentence to the discussion in Section 8 (text in blue below) Finally, we note that the IPA in the numerical studies was readily calculated for the conditional survival function as you suggested. We have now made this clearer by updating the corresponding paragraph in Section 6.

> *Quote from revised manuscript, Changes in Section 5*
>
> The norm defined in equation (13) operates on functions F which are features of the observed data distribution. This means that Proposition 2 and Corollary 3 provide guarantees in terms of a how well the function $\hat{\varphi}_n$ predicts the observed data. Ideally, we would like performance guarantees for the estimated cumulative hazard functions $\hat{\Lambda}_{jn}$ or for the risk predictions of the event of interest $\hat{Q}_n$ defined in equation (11). For this we note the one-to-one correspondence between the learner $\hat{\varphi}_n$ and the tuple of learners $(\hat{\Lambda}_{1n}, \hat{\Lambda}_{2n}, \hat{\Gamma}_n)$ established in equations (3)-(4) and (7). Based on this correspondence we postulate that the performance guarantees provided for $\hat{\varphi}_n$ will in many cases translate into corresponding performance guarantees for elements of the tuple $(\hat{\Lambda}_{1n}, \hat{\Lambda}_{2n}, \hat{\Gamma}_n)$ and also to derived measures such as the risk of the event of interest. We do not analyse this hypothesis further theoretically, but investigate the performance of the joint survival super learner for predicting the risk of the event of interest in our numerical experiments in Section 6.

Quote from revised manuscript, Changes in Discussion

A potential drawback of our approach is that we are evaluating the loss of the learners on the level of the observed data distribution, while the target of the analysis is either the event-time distribution, or the censoring distribution, or both. However, our numerical experiments show that our approach can perform well for estimating the target of the analysis.

Quote from revised manuscript, Changes in Section 6

Each super learner provides a learner for the cumulative hazard function for the outcome of interest. From the cumulative hazard functions, we obtain a predicted risk of the combined outcome event, c.f., equation (11), where in the current scenario without competing risks, $\Lambda_2 = 0$, and $\hat{Q}_n(T \leq t, D = 1 \mid X = x) = 1 - \exp\{-\hat{\Lambda}_{1n}(t \mid x)$. We measure the performance of the risk prediction models by calculating the index of prediction accuracy (IPA) [Kattan and Gerds, 2018] at a fixed time horizon (36 months). For a risk prediction model $r\colon \mathcal{X} \to [0, 1]$, IPA at time τ is defined as

$$1 - \frac{\mathbb{E}_Q[(r(X) - \mathbb{1}\{T \leq \tau\})^2]}{\mathbb{E}_Q[(Q(T \leq \tau) - \mathbb{1}\{T \leq \tau\})^2]}.$$

We chose IPA as a performance measure because it is proper, incorporates both discrimination and calibration, and is easy to interpret as it measures the relative performance gain compared to the null model which does not use any covariate information. The definition of IPA involves the uncensored survival time T, which is not available in practice. However, in the numerical studies, this quantity is available because we know the data-generating mechanism. For the presentation of results we estimate the IPA in a large independent data set of uncensored survival times ($n = 20,000$).

1. Second, and more minor, I wonder if the authors could expand on why it is difficult to extend the method to an ensemble estimator in this setting.

   Thank you for this comment. We do not believe that it needs to be particularly difficult, but it is beyond the scope of our manuscript. We have now rephrased and expanded this paragraph in the discussion.

Quote from revised manuscript

We have focused on a discrete version of the joint survival super learner, but it is straightforward to extend the method to more complex ensemble learning, e.g., through stacking [Breiman, 1996]. However, there are multiple possible ways to apply stacking to the the joint survival super leaner. One option is to construct a single convex combination of the F-learners $\varphi \in \Phi(\mathcal{A}_1, \mathcal{A}_2, \mathcal{B})$. Another, perhaps more interesting option, is to construct three separate convex combinations of the learners in $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{B}$. How such an ensemble should be build and implemented is an interesting topic for future research.

Reviewer 2

Section 3:

- P5. The point that the partial log-likelihood does not work well as an evaluation criterion is interesting and warrants further elaboration. An additional sentence explaining this point would be helpful. Additionally, I wonder if this problem can be circumvented in other ways: perhaps models that normally yield piecewise constant hazards could be included in a slightly modified form with smoothed hazards. What would be the consequences of this approach?

  We agree that this is an important point, and we have elaborated further. It is an interesting idea to attempt to smooth the piecewise constant cumulative hazard functions. However, we believe that this introduces other issues that would have to be addressed, and we briefly reflect on that now. Please see the updated paragraph below with new text in blue.

Quote from revised manuscript

However, the partial log-likelihood loss does not work well as a general purpose measure of performance in hold-out samples when data are observed in continuous time. The reason is that the partial log-likelihood assigns an infinite value to any learner that predicts piecewise constant cumulative hazard functions, if the test set contains event times that are not observed in the training set. For instance, if no competing risks are present, a piecewise constant cumulative hazard function postulates a model for the distribution of the survival times where all probability mass is assigned to the finite number of time points at which the cumulative hazard function jumps. The likelihood according to such a model is zero at almost all time points, and thus the likelihood of any hold-out sample will almost surely be zero when data are observed in continuous time. This problem occurs with prominent survival learners including the Kaplan-Meier estimator, random survival forests, and semi-parametric Cox regression models, and these learners cannot be included in the library of the super learner proposed by Polley and van der Laan (2011). A reviewer suggested that one might approach this issue by post-hoc smoothing of the piecewise constant cumulative hazard functions. This is an interesting idea which we however do not pursue further in this work.

Section 4:

- P5. I find the notation using "1" for the indicator function difficult to read, especially when preceded by other numerals. Why not use I, δ, or a bold or blackboard/double-struck 1?

  Thank you for this suggestion. We now use the blackboard/double-struck 1.

- P6. The manuscript explains that it is in principle fairly easy to use the survival package to estimate libraries of models for A_1, A_2, and B. The practical value of the article would improve with a web appendix showing code implementing this for the prostate cancer study data.

  Thank you for this suggestion. We refer to our accompanying github repo:
  `https://github.com/amnudn/joint-survival-super-learner`
  where we have now added an example that demonstrates how the survival package can be used to construct learners. We cannot share the original data, but we use an emulated data set to demonstrate the application of the joint survival super learner.

Section 5:

- P8. You state that a sensible objective function must be proper. While I don't disagree, this point often generates discussion. One might argue that in certain problems, our objective need not be the correct recovery of the probability distribution. Rather, we may want to predict outcomes at specific time points while incorporating relative costs of incorrect predictions. It's not clear that using probabilities as an intermediate step is always optimal, especially when these are estimated with error. Furthermore, while a proper loss function might be optimal with large sample sizes, this may not hold for finite samples. More discussion of this point would be valuable.

  We thank the reviewer for this thoughtful comment. We agree that in some applications, the goal is not to estimate the true underlying risk or probability distribution, but rather to develop a decision model that directly minimizes expected loss under specified costs of incorrect predictions.

  However, in our experience it is rare in medical settings that subject matter experts are able to explicitly quantify the costs and benefits of decisions that are based on predicted risks of future events, in particular, in competing risk settings where decisions likely have to consider all risks simultaneously.

  Our work, by contrast, focuses on super learning of risks – that is, prediction of outcome probabilities – and particularly on simultaneous estimation of the censoring distribution, rather than relying on a prespecified but potentially misspecified regression model for the censoring distribution. A detailed discussion of decision-analytic trade-offs or the distinction between correct and incorrect categorical predictions would thus be outside the scope of our paper.

  Finally, we maintain that the objective (or loss) function should be proper even in small samples. Improper loss functions can lead to misleading conclusions about model performance, as illustrated for example by the integrated discrimination improvement (IDI) and net reclassification improvement (NRI) metrics (see Hilden and Gerds [2014]), and this happens in small samples as well as in large samples.

- P8. The authors quickly move from using proper scoring rules to the Brier score, which feels less general. Are the results expected to hold for other proper scoring rules?

  Yes, they probably would. But there are few alternative strictly proper scoring rules for our use case: Savage proved in 1971 [Savage, 1971] that in case of predicted probabilities there are only two strictly proper scoring rules: the Brier score and the logarithmic score. We work with the Brier score because it readily works for tupels of predicted probabilites and does not have issues when predicted probabilites are exactly zero or one. It

may be possible however to define integrated versions of the logarithmic score and we would expect similar results.

- P9. At the top of the page, it's argued that the oracle inequality provides insights into how the number of folds, time horizon, and number of learners influence performance. Could the practical utility of this be illustrated in the context of the prostate cancer example?

  Indeed, our statement was misleading. We do not think that that actual practical guidance can be derived easily from the inequality. We have now deleted this statement:

  > *Quote from revised manuscript*
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  > As with many finite-sample oracle inequalities, this result is of little direct practical utility because the right-hand side depends on data-dependent, unknown quantities. ~~However, it does quantify how the number of folds, the time horizon, and the number of learners in the library can be expected to influence the performance.~~

- P10. The performance will be specified using the IPA, which needs more discussion. Please briefly state how it is defined and why it was chosen. (Is the IPA itself proper? If not, does this still make sense here? Do we need other metrics as well?)

  Thank you for pointing this lack of clarity out. We have now defined the IPA explicitly, please see our response to Reviewer 1's first query. The IPA is a scaled version of the Brier score, where the scaling does not depend on the model, and hence it is indeed proper.

- P10. The SurvSL model should be described in more detail.

  Agreed. We have expanded this section as follows:

Quote from revised manuscript

For the second aim, we consider the super learner *survSL* proposed by [Westling et al., 2021] which like the joint survival super learner does not require a pre-specified censoring model. The super learner *survSL* also implements an ensemble learning approach, combining multiple learners to improve predictive performance. The major difference is that *survSL* iterates between prediction of the outcome probabilities and prediction of the censoring probabilities, where the joint survival super learner evaluates all learners simultaneously on the level of the observed data distribution. Another difference is that *survSL* creates a meta-learner that combines the predictions of the individual learners, whereas the joint survival super learner applied in the simulation study is a discrete super learner. Since both approaches provide estimates of the event-time and the censoring distributions, we compare their performance with respect to both. Again we use the IPA to quantify the predictive performance.

- P10. For the "second aim," it's suggested that an advantage of the joint survival learner may be that it is a discrete super learner. This warrants more explanation – why is this an advantage? Is there more risk of overfitting in small samples?

  This is an interesting point. To our knowledge, the finite sample performance of a discrete versus continuous (ensemble) super learner is not well understood. We conjecture that the discrete super learner may be advantageous in the case where at least one of the learners in the library is consistent and the data set is large enough, whereas the ensemble super learner could have an advantage when all learners are misspecified to some extent. We have expanded on this and tried to clarify that this only a possible explanation and not a strict fact.

> *Quote from revised manuscript*
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> We see that for most sample sizes, the joint survival super learner has similar or higher IPA compared to *survSL* with respect to both the prediction of the censoring and the outcome risks. We note that the advantage of the joint survival super learner in this particular simulation setting might be due that it is a discrete super learner, whereas *survSL* combines the learners. We conjecture that a discrete super learner may be advantageous in the case where at least one of the learners in the library is consistent and has converged given the available data set, whereas the ensemble super learner could have an advantage when all learners are misspecified to some extent, or have not converged sufficiently (in different regions of the covariate space). In this simulation study, the data generation model is a low-dimensional parametric model which was included in the library of learners and hence this could be an explanation of why the discrete joint survival super learner outperforms *survSL* in this scenario.

Section 7:

- P11. The prostate cancer study seems like an afterthought. It would be beneficial to reference it more when introducing the problem to demonstrate practical relevance.

  Great point. We have now added the following sentence to the introduction:

  > *Quote from revised manuscript*
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  > We demonstrate how to construct a library of learners using common methods for survival analysis and illustrate the use of the joint survival super learner using data from a prostate cancer study Kattan et al. [2000] and also synthesize the prostate cancer data to generate a realistic scenario for our simulation experiments.

- P11. The splitting into training and test data seems wasteful. Could nested cross-validation be used instead?

  There is a trade-off here between the performance of an average model that can be build with a given sub-sample size of say 90% of the data, corresponding to 10-fold cross-validation, and the performance of an actual model that was built with a given training sample. Figure 4 of our manuscript shows the actual predictions of one model in the single-split test sample. If we would switch to 10-fold cross-validation then the predicted probabilities would be coming from 10 different models. This may as well be of interest but it shows something else and is in our

opinion not generally preferable when the task is to illustrate a new method.

- P11. More discussion of the results is needed. What exactly do we learn from this practical use case?

  Good point. We have expanded as follows.

  > *Quote from revised manuscript*
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  > Figure 4 compares the 3-year risk predictions from the pre-specified Cox model and the discrete joint survival super learner in the test set. The figure shows that the differences between the conventional method and our new method are mainly due to differences in the estimated censoring probabilities, while the event probabilities remain similar.
  > These results illustrate that the joint survival super learner identified different learners for the cause-specific event and censoring processes, demonstrating its ability to adapt flexibly to complex data structures. In this application, the difference in predictive accuracy over pre-specified Cox models seems to be modest, indicating that for this particular dataset the main advantage of the joint survival super learner lies in its data-adaptive model selection rather than in dramatic performance improvement. Importantly, the consistent selection of the random survival forest for the censoring distribution shows that accounting for informative censoring can be learned automatically rather than being fixed a priori.

- P11. The usefulness of this manuscript for practical researchers would be greatly enhanced with a link to a code supplement.

  Yes, we share our codes via a github repository. Here is what we state at the end of the introduction

  > *Quote from revised manuscript*
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  > Code and an implementation of the joint survival super learner in R [R Core Team, 2024] are available at `https://github.com/amnudn/joint-survival-super-learner`.

Section 8:

- P12. The stated drawback—that the authors evaluate the loss of learners at the level of the observed data distribution while the target is either the event-time distribution, censoring distribution, or both—needs more detailed explanation. Are there alternatives to this approach, and what would be their drawbacks?

We have addressed the first part of your point in our answer to Reviewer 1's first point, please see above. Regarding alternative methods and their drawbacks, we do mention alternative approaches in Section 3, and we now reiterate and expand on these points in the Discussion, please see the paragraph below which we have added to the Discussion.

> *Quote from revised manuscript*
>
> A major advantage of the joint survival super learner is that the performance of each combination of learners can be estimated without additional nuisance parameters. A potential drawback of our approach is that we are evaluating the loss of the learners on the level of the observed data distribution, while the target of the analysis is either the event-time distribution, or the censoring distribution, or both. However, our numerical experiments show that our approach can perform well for estimating the target of the analysis. Alternatives to our approach including IPCW loss, censoring unbiased transformations, and pseudo-values, were discussed in Section 3. As mentioned, the drawback of these approaches is that they all need a pre-specified estimator of the censoring distribution, and hence these methods are not immediately applicable if we do not in advance know how to model the censoring distribution. Another alternative approach to ours is the strategy underlying the super learner *survSL* [Westling et al., 2021]. Though we found this approach to perform reasonably well in our simulation study, the approach lacks formal theoretical guarantees and has not been extended to competing risks settings. We note that the partial log-likelihood loss, like our suggested approach, also measures performance with respect to a feature defined by the observed data distribution [e.g., Hjort, 1992, Whitney et al., 2019].

- P12. The authors claim that targeted learning is also known as debiased machine learning. However, these are often presented as distinct approaches. This should be clarified.

  This is a fair point. We have updated the text and replaced "known as" with "or" as follows:

Quote from revised manuscript

A relevant application of the joint survival super learner is within the framework of targeted learning [van der Laan and Rose, 2011] or debiased machine learning [Chernozhukov et al., 2018], which are general methodologies for combining flexible, data-adaptive estimation of nuisance parameters with asymptotically valid inference for low-dimensional target parameters.

- P12. The discussion lacks reflection on the results from sections 6 and 7.

  We have added the following paragraph at the very end of the discussion.

Quote from revised manuscript

The findings from our simulation study suggest that the joint super learning framework can be used to construct a well-performing risk prediction model without a pre-specified estimator of the censoring distribution. Our analysis of the prostate cancer data demonstrate the practical feasibility of the joint survival super learner. In particular, it shows that our super learner was able to automatically identify a complex model for the censoring distribution in a setting where the censoring distribution was expected to be highly dependent on baseline covariates.

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